

Neutrophil extracellular traps enriched in oxidized mitochondrial DNA are interferogenic and contribute to lupus-like disease

SCIFACT-EVIDENCE-1049501

Evidence abstract

Neutrophil extracellular traps (NETs) are implicated in autoimmunity, but how they are generated and their roles in sterile inflammation remain unclear.

Ribonucleoprotein immune complexes (RNP ICs), inducers of NETosis, require mitochondrial reactive oxygen species (ROS) for maximal NET stimulation.

After RNP IC stimulation of neutrophils, mitochondria become hypopolarized and translocate to the cell surface.

Extracellular release of oxidized mitochondrial DNA is proinflammatory in vitro, and when this DNA is injected into mice, it stimulates type I interferon (IFN) signaling through a pathway dependent on the DNA sensor STING.

Mitochondrial ROS are also necessary for spontaneous NETosis of low-density granulocytes from individuals with systemic lupus erythematosus.

This was also observed in individuals with chronic granulomatous disease, who lack NADPH oxidase activity but still develop autoimmunity and type I IFN signatures.

Mitochondrial ROS inhibition in vivo reduces disease severity and type I IFN responses in a mouse model of lupus.

Together, these findings highlight a role for mitochondria in the generation not only of NETs but also of pro-inflammatory oxidized mitochondrial DNA in autoimmune diseases.

Document metadata

Corpus | SciFact

Document ID | 1049501